

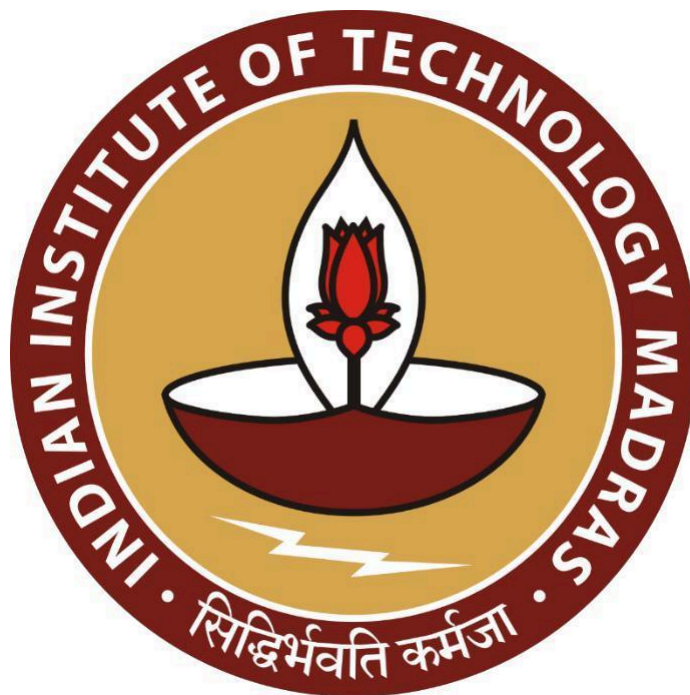
# **Optimizing Retail Efficiency: A Data-Driven Approach for LAXMI GENERAL STORE**

**A Proposal report for the BDM capstone Project**

Submitted by

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## **Declaration Statement**

I am working on a Project titled “**Optimizing Retail Efficiency**”. I extend my appreciation to Laxmi General Store, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

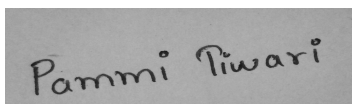
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

A rectangular box containing a handwritten signature in black ink that reads "Pammi Tiwari".

Name: Pammi Tiwari

Date: 05/08/2024

## **Executive Summary**

Laxmi General Store is a small B2C grocery store in Preetam Nagar, Prayagraj, established in 2021 and owned by Shivam Bhartiya. The store caters to the local community's daily needs with a wide range of products, including groceries, household essentials, personal care items, and packaged foods. Known for quality products at competitive prices, the store ensures customer satisfaction and loyalty.

The store faces low profit margins due to poor inventory management. Frequently, high-demand items like rice, flour, cooking oils, and toiletries run out of stock, causing customer dissatisfaction. Conversely, low-demand items like niche cleaning products and certain packaged foods lead to excess inventory and increased holding costs. This results in tied-up capital and potential losses from unsold goods.

To address these issues, we will analyze sales data to understand demand patterns, ensuring high-demand items are always in stock and reducing excess low-demand products. By optimizing ordering practices and strategically placing products, we can minimize stockouts and overstocking, freeing up capital, improving cash flow, and increasing profitability. These measures will enhance operational efficiency, reduce costs, and provide a better shopping experience for customers.

## **Organization Background**

Laxmi General Store is a neighborhood grocery store located in Preetam Nagar, Prayagraj. Established in 2021 by Shivam Bhartiya, the store has become a trusted source for everyday household needs. It operates within the B2C sector, directly serving the local community with a wide variety of products. The store's inventory includes groceries, fresh produce, packaged foods, household essentials, and personal care items.

Laxmi General Store prides itself on offering high-quality products at affordable prices, ensuring that customers receive value for their money. The store is well-known for its friendly customer service and its commitment to meeting the needs of the residents in the area. Despite its success, the store faces challenges with inventory management, which affects its profitability and customer satisfaction. By addressing these issues, Laxmi General Store aims to improve its operations and continue to serve the community effectively.

## **Problem Statement**

1. **Unclear Spending Patterns:** Laxmi General Store lacks a clear understanding of customer spending habits across different categories like groceries, toiletries, cleaning supplies, and personal care items. Without proper analysis of past sales data, optimizing inventory to better serve customers and maximize profits is challenging.
2. **Stockout Woes:** Frequent stockouts of popular items such as rice, flour, and daily-use toiletries frustrate customers and lead to lost sales. The store struggles to predict when customers will need these items again, resulting in unsatisfied customers who may look elsewhere for their needs.
3. **Budgeting Blindspot:** Allocating the store's budget effectively is difficult without a detailed breakdown of spending across various product categories. This lack of insight can lead to overspending on some items and understocking others, which negatively impacts customer satisfaction and the store's overall profitability.

## **Background of the Problem**

### **Internal Problems:**

1. **Lack of Spending Insight:** Laxmi General Store struggles to analyze customer spending across categories like groceries, toiletries, and personal care items, complicating inventory decisions.
2. **Frequent Stockouts:** Popular items like rice, flour, and toiletries often run out, leading to frustrated customers and lost sales.
3. **Overstocking of Low-Demand Items:** Excess inventory of niche cleaning products and certain packaged foods ties up capital and increases the risk of expired goods.
4. **Ineffective Budget Allocation:** Without detailed spending breakdowns, the store overspends on some items and understocks others, affecting profitability.

### **External Problems:**

1. **Supplier Reliability:** Inconsistent deliveries and supply chain issues cause stock outs or overstock situations, disrupting inventory levels.
2. **Market Competition:** Nearby competitors with similar products and competitive prices pressure the store to manage inventory effectively to avoid losing market share.

3. **Economic Factors:** Economic fluctuations, such as changes in consumer purchasing power and inflation, make demand forecasting difficult and impact inventory management.

## **Problem Solving Approach**

1. **Data Collection and Rapport Building:**

To ensure accurate data collection and analysis, we will start by fostering a collaborative environment with the staff at Laxmi General Store. Engaging with employees and explaining the project's benefits will secure their buy-in and facilitate a smooth data collection process. A meeting with the store owner, Shivam Bhartiya, will define the project objectives, data requirements, and collection methods. This foundational step is crucial for the success of the subsequent phases.

2. **Analyzing Historical Sales Data:**

Using Excel or Google Sheets, we will organize and clean the provided historical sales data. This will involve sorting, filtering, and categorizing sales data into groceries, toiletries, cleaning supplies, and personal care items. By understanding which categories generate the highest revenue and profit margins, we can make informed decisions about inventory management.

3. **Demand Forecasting and Inventory Optimization:**

Based on the sales data analysis, we will use Python with libraries such as Pandas, Scikit-learn, and Statsmodels to implement demand forecasting techniques. These techniques, including time series analysis, will help predict future sales for different product categories. This will ensure that high-demand items are always in stock while reducing excess inventory of low-demand products. We will establish a reorder point system to automate the replenishment process, minimizing stockouts and overstocking.

4. **Supplier Collaboration:**

Partnering with local suppliers and leveraging their expertise can enhance product quality and streamline the supply chain. By analyzing product origin data from sales, we can identify items that can be sourced locally, reducing transportation times and costs. Strengthening relationships with suppliers will facilitate more efficient ordering and delivery processes, leading to cost reductions and minimized waste.

5. **Strategic Product Placement and Marketing:**

Using insights from the sales data analysis, we will strategically place high-demand items in prominent locations within the store to boost visibility and sales. Additionally, targeted marketing campaigns will be designed to promote these items.

Low-performing categories will be further analyzed to identify specific underselling items, which may lead to adjustments such as bundled promotions or recipe reformulation.

**6. Customer Segmentation and Targeted Promotions:**

By applying clustering techniques to the purchase data using Python, we will segment customers based on their buying habits and preferences. These segments will be targeted with specific marketing campaigns, promotions, and loyalty programs that cater to their unique needs. This data-driven approach will significantly improve customer engagement and retention.

**7. Continuous Monitoring and Improvement:**

We will use data visualization tools like Tableau, Power BI, and Python libraries such as Matplotlib and Seaborn to set up simple metrics to track inventory turnover, stockout rates, and holding costs. Regularly checking these metrics will help us see how well the new inventory strategies are working. We will also ask for feedback from customers and staff to find more ways to improve. This will ensure that the store's operations stay efficient and meet customer needs.

By implementing these problem-solving approaches, Laxmi General Store aims to optimize its inventory management, improve customer satisfaction, and increase profitability.

**Analytical Tools**

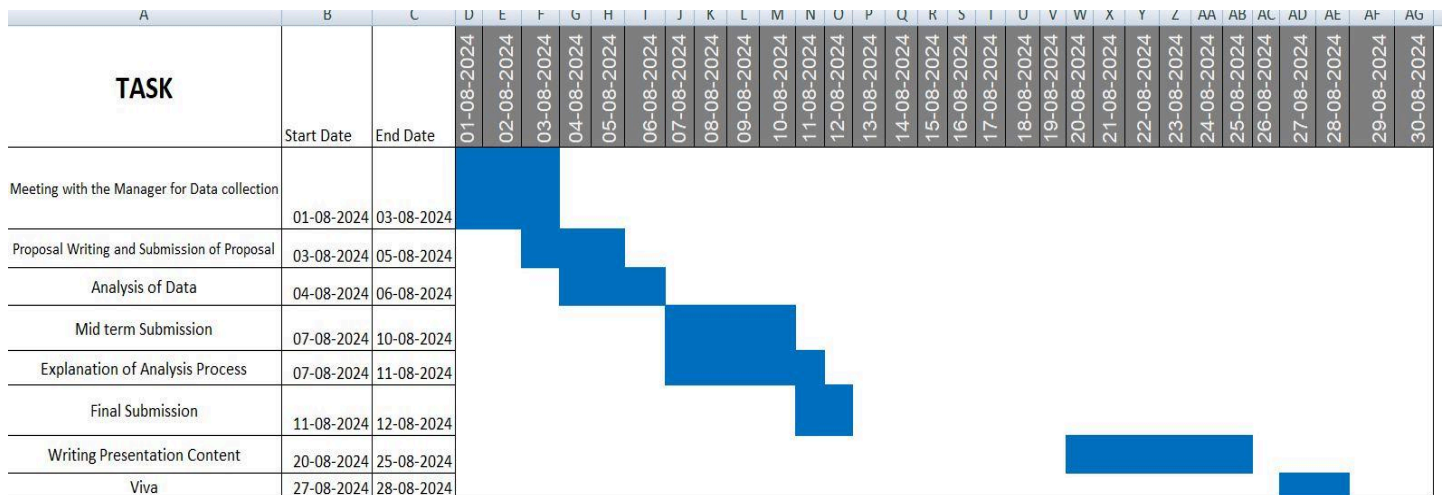
1. Excel or Google Sheets:.
2. Python:
3. Data Visualization

**Expected Timeline:**

| S.No | Activity                                     | Start Date | End Date   |
|------|----------------------------------------------|------------|------------|
| 1    | Meeting with the Manager for Data collection | 01/08/2024 | 03/08/2024 |
| 2    | Proposal Writing and Submission of Proposal  | 03/08/2024 | 05/08/2024 |
| 3    | Analysis of Data                             | 04/08/2024 | 06/08/2024 |

|   |                                 |            |            |
|---|---------------------------------|------------|------------|
| 4 | Mid term Submission             | 07/08/2024 | 10/08/2024 |
| 5 | Explanation of Analysis Process | 07/08/2024 | 11/08/2024 |
| 6 | Final Submission                | 11/08/2024 | 12/08/2024 |
| 7 | Writing Presentation Content    | 20/08/2024 | 25/08/2024 |
| 8 | Viva                            | 27/08/2024 | 28/08/2024 |

## GANTT CHART



## **Expected Outcome**

The implementation of the proposed inventory management and data analysis solutions at Laxmi General Store is expected to bring significant improvements:

### **Reduced Stockouts:**

1. **Enhanced Customer Satisfaction:** Preventing stock outs of high-demand items ensures customers find what they need, increasing satisfaction and loyalty.
2. **Increased Sales:** Consistently stocking popular items reduces missed sales opportunities, boosting overall revenue.

### **Optimized Inventory:**

1. **Balanced Stock Levels:** Identifying understocked and overstocked categories minimizes waste and ensures adequate stock of fast-moving items.
2. **Cost Efficiency:** Maintaining optimal inventory levels reduces holding costs and financial strain from excess inventory or emergency restocking.

### **Targeted Marketing:**

1. **Customer Segmentation:** Analyzing purchase data allows for tailored promotions, increasing marketing effectiveness.
2. **Higher Engagement:** Personalized campaigns engage customers better, leading to increased sales from promotions.

### **Strategic Product Placement:**

1. **Enhanced Visibility:** Placing high-demand items in prominent locations boosts visibility and encourages impulse purchases.
2. **Sales Growth:** Strategic placement of high-performing products can increase sales and revenue.

### **Improved Product Quality:**

1. **Fresher Ingredients:** Partnering with local suppliers ensures fresher produce, enhancing product quality.
2. **Customer Trust:** Higher product quality builds trust and satisfaction, encouraging repeat business.

### **Streamlined Supply Chain:**

1. **Efficient Processes:** Closer supplier relationships streamline ordering and delivery, reducing stockouts and delays.
2. **Cost Reductions:** Optimized supply chain operations lead to cost savings through bulk purchasing and reduced transportation expenses.

### **Reduced Waste:**

1. **Minimized Overstocking:** Improved supplier communication and shorter lead times enable accurate ordering, reducing overstock.
2. **Decreased Spoilage:** Better inventory management minimizes spoilage of perishable items, reducing waste.